

Phase-State Projection (PSP) Model of Relativistic Reality

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1. Fundamental Definitions

1.1 Universal Informational Field

Define a universal complex-valued informational field:

$$\Psi : M^4 \times \mathbb{R}^n \rightarrow \mathbb{C}, \quad \Psi = \Psi(x^\mu, \phi_j)$$

where:

- $x^\mu = (t, x, y, z) \in M^4$: spacetime coordinates
- $\phi_j \in \mathbb{R}^n$: phase-state dimensions (kinematic, quantum, thermodynamic, etc.)

1.2 Observer Phase-State Vector

Each observer is characterized by a time-dependent phase-state:

$$\Phi_{\text{obs}}(t) = \begin{bmatrix} \phi_1(t) \\ \phi_2(t) \\ \vdots \\ \phi_n(t) \end{bmatrix}$$

with kinematic phase component:

$$\phi_{\text{kin}}(v) = \phi_{\text{kin}}^{(0)} + \int_0^v \beta(u) du, \quad \text{with } \beta(u) \approx \frac{u}{c^2}$$

1.3 Reality Projection Operator

Observed reality is defined as a projection:

$$R(x^\mu) = P[\Psi, \Phi_{\text{obs}}] = \int_{\mathbb{R}^n} \Psi(x^\mu, \phi) \cdot K(\phi | \Phi_{\text{obs}}(t)) d^n \phi$$

with projection kernel:

$$K(\phi | \Phi) = \exp \left(i \sum_{j=1}^n \phi_j \cdot F_j(x^\mu) \right)$$

2. Dynamics of Field

2.1 Action

$$S[\Psi] = \int d^4x d^n \phi \left[\frac{1}{2} |\partial_\mu \Psi|^2 - \frac{1}{2} |\partial_{\phi_j} \Psi|^2 - V(\Psi) \right]$$

2.2 Field Equation

$$\square\Psi - \nabla_\phi^2\Psi + \frac{\delta V}{\delta\Psi^*} = 0$$

where:

$$\square = \partial^\mu\partial_\mu, \quad \nabla_\phi^2 = \sum_{j=1}^n \frac{\partial^2}{\partial\phi_j^2}$$

3. Modified Simultaneity Relation

For an observer moving with velocity v , their effective simultaneity time for distant location at position $\vec{r}$ is:

$$t'_{\text{effective}} = t_0 - \frac{\vec{v} \cdot \vec{r}}{c^2} - \frac{\Delta\phi_{\text{kin}}}{\omega_0}$$

with phase shift:

$$\Delta\phi_{\text{kin}} = \phi_{\text{kin}}(v) - \phi_{\text{kin}}^{(0)}$$

4. Quantum Correspondence

The standard quantum wavefunction is recovered as a marginal projection:

$$\psi(x^\mu) = \int \Psi(x^\mu, \phi) \cdot \rho(\phi) d^n\phi$$

where $\rho(\phi)$ is the observer-specific phase distribution.

5. Constraints on Projection Freedom

To preserve objective reality and prevent arbitrary projection construction:

$$\left| \frac{d\phi_{\text{kin}}}{dt} \right| \leq \omega_c \quad (\text{phase coherence bandwidth})$$

$$|\nabla_\phi R(x^\mu)| \leq \Lambda_{\text{coh}} \quad (\text{causal projection constraint})$$

6. Experimental Prediction

Apparent simultaneity shift for an observer moving at velocity v toward a galaxy at distance d :

$$\Delta t_{\text{apparent}} = -\frac{vd}{c^2} - \frac{\Delta\phi_{\text{kin}}}{\omega_0}$$

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